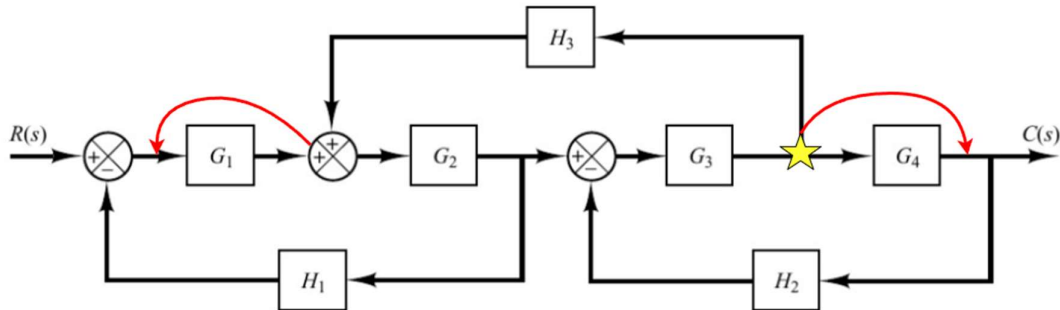
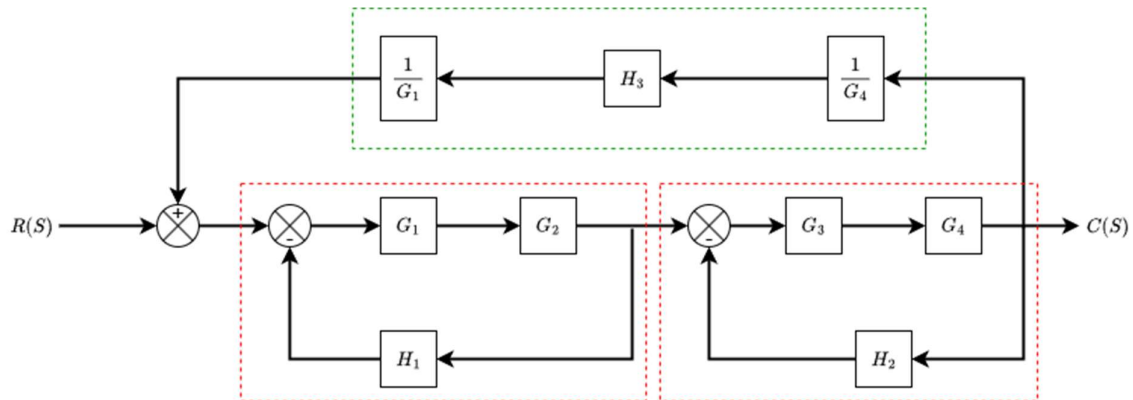


Practice Problems 1 Solutions: Block Diagram Algebra

1)

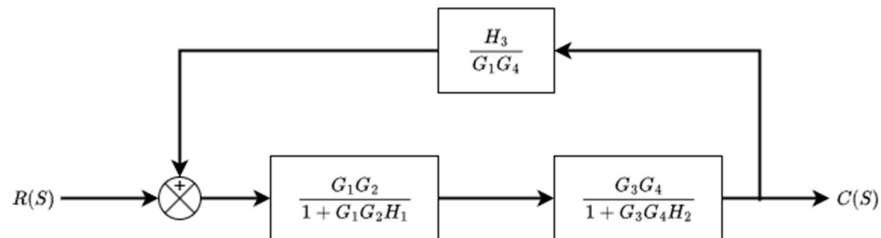


To simplify this block diagram, the take-off point between G_3 and G_4 can be moved to the right end, and the summing junction between G_1 and G_2 can be moved to the left end, resulting in the block diagram below.



Note: the order of the summing junctions does not matter, so they have been swapped for clarity.

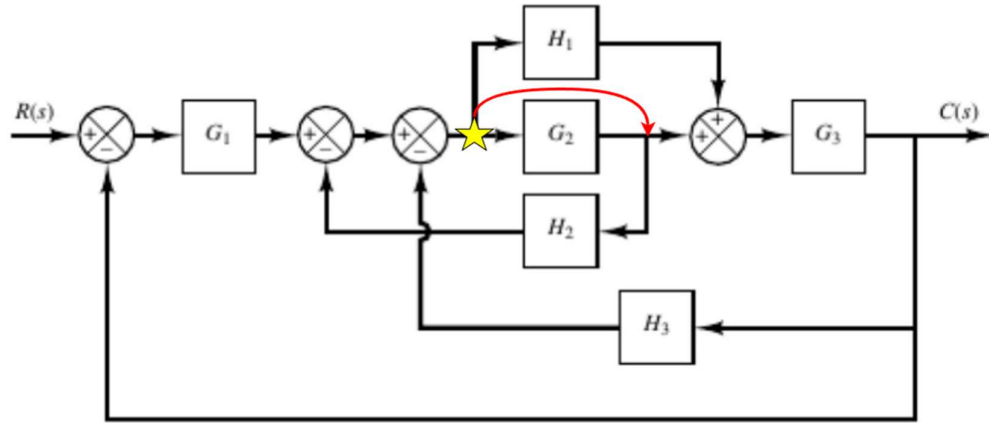
This creates two cascaded feedback loops (in red) and three cascaded transfer functions (in green). Each can be simplified to get the block diagram below.



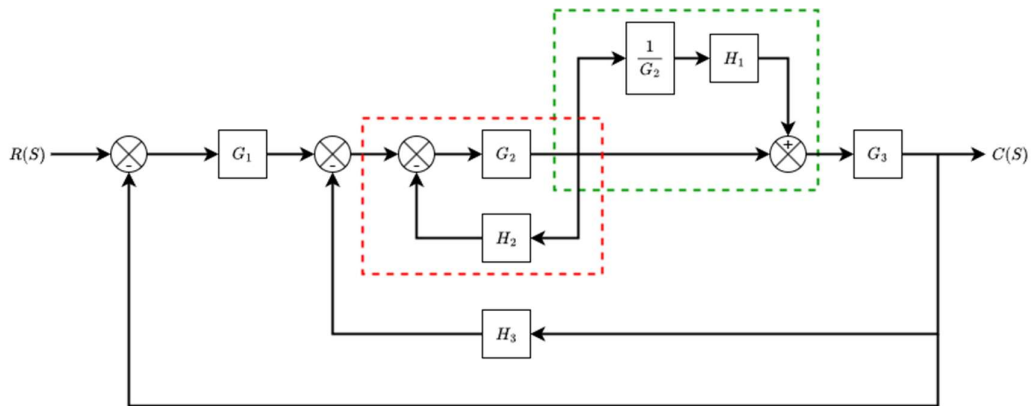
This creates a positive feedback loop, which can be simplified to arrive at the final transfer function below.

$$\frac{G_1 G_2 G_3 G_4}{(1 + G_1 G_2 H_1)(1 + G_3 G_4 H_2) - G_2 G_3 H_3}$$

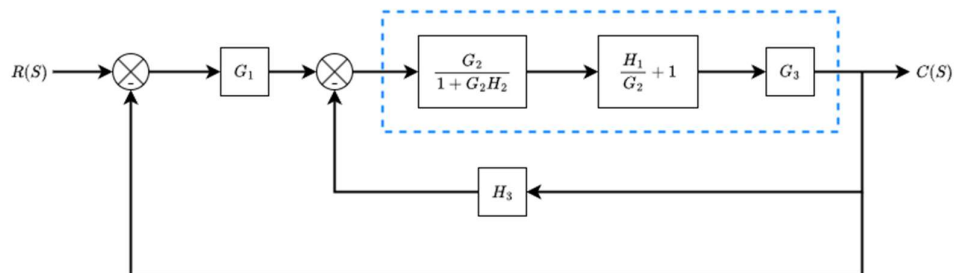
2)



To simplify this block diagram, the take-off point leading to H_1 , can be moved to the right of G_2 .



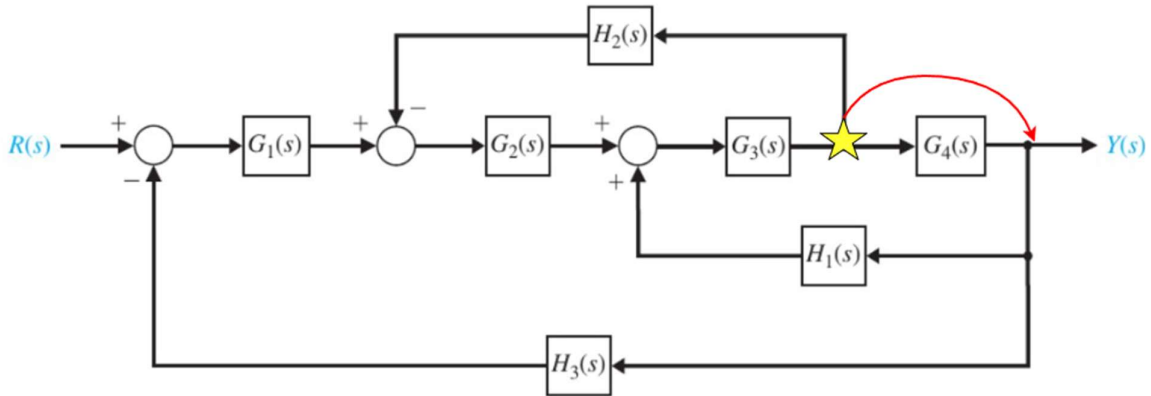
This creates a negative feedback loop (in red) and a parallel connection (in green), which can be simplified as shown below.



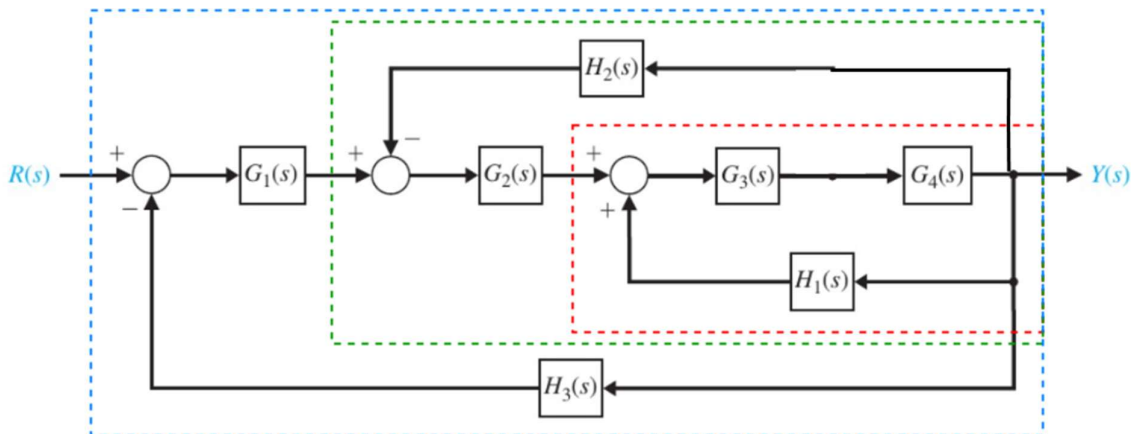
Once the cascaded transfer functions (in blue) are combined into a single transfer function, a nested loop is formed. This can be simplified to obtain the final transfer function below.

$$\frac{G_1 G_3 (G_2 + H_1)}{1 + G_2 H_2 + G_3 (G_1 + H_3) (G_2 + H_1)}$$

3)



To simplify this block diagram, the take-off point between G_3 and G_4 can be moved to the right end.



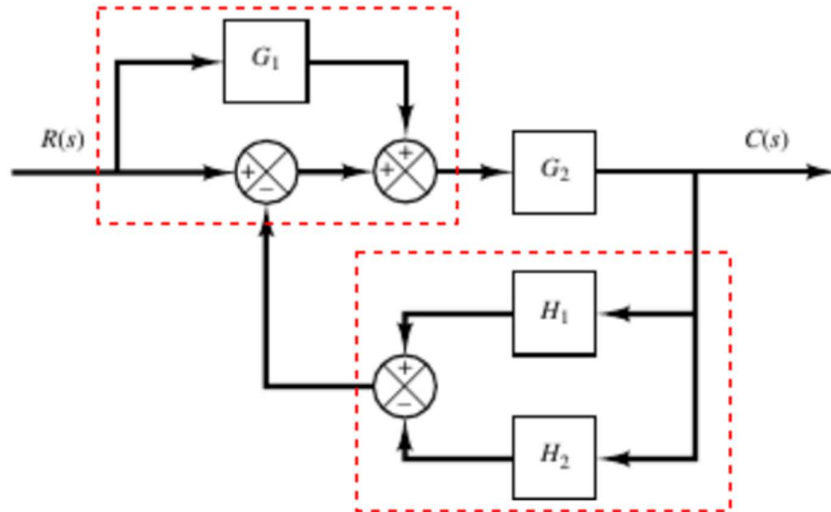
This creates three nested loops:

- An innermost positive feedback loop (in red)
- An outer negative feedback loop (in green).
- An outermost negative feedback loop (in blue).

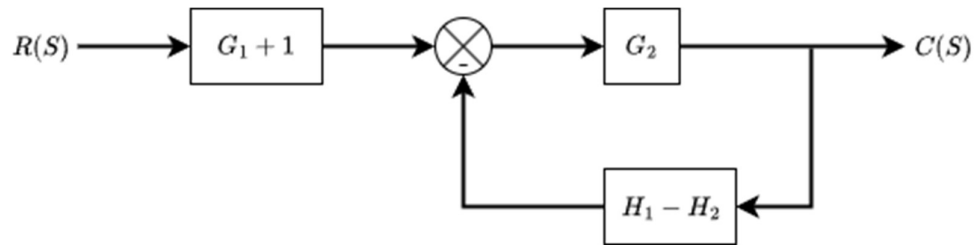
These can be simplified from the inside out to arrive at the final transfer function below.

$$\frac{G_1 G_2 G_3 G_4}{(1 - G_3 G_4 H_1 + G_2 G_3 H_2) + (G_1 G_2 G_3 G_4 H_3)}$$

4)



There are two parallel connections (in red), which can be simplified as shown in the block diagram below.



This creates a negative feedback loop cascaded with $G_1 + 1$, which can be simplified to obtain the final transfer function below.

$$\frac{G_2(G_1 + 1)}{1 + G_2(H_1 - H_2)}$$